

Exam Prep Semester 1

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Single Variable Calculus

Exam Layout

3 hours long, 8 questions

Questions are similar to tutorial and homework problems

Content

For a list of all content see Content.pdf

Tutorial Questions

1.
 - Calculating domain and range
 - **Showing bijectivity**
 - Using rational root theorem and polynomial division
 - Trigonometric addition formulae
2.
 - Calculating inverse functions
 - Simplifying trig. expression with inverses in them
 - **Computing limits using epsilon-delta definition**
 - Computing limits
 - Determining when limits exist
3.
 - Proving roots exist using Bolzano's theorem
 - Determining when functions are continuous
 - Computing limits at infinity/ when denominator is zero
 - Using definition of derivative
 - **Proving positivity at nearby points**
4.
 - Computing derivatives by the rules
 - Logarithmic differentiation
 - Implicit differentiation
5.
 - Using Rolle's theorem with Bolzano's
 - **Using the mean value theorem to approximate values**
 - Using L'hospital's rule to compute indeterminate limits
 - Find values for which a function is increasing/ decreasing
6.
 - Find values for which a function is concave up/ down
 - Computing local minima/ maxima
 - Computing global minima/ maxima on a closed interval
 - Finding linear approximations to a function at a point
 - Related rates of change
 - Optimisation problems
 - Estimating roots using Newton's method
7.
 - Compute area under a curve
 - Compute volume of a curve rotated
 - Compute arc lengths of a graph
 - Show a function is integrable using Riemann Sums
 - Compute integrals using integration by parts and substitution
8.
 - Computing integrals of powers of sine and cosine, tangent and cosecant
 - Compute volume of a curve rotated
 - Integration by parts
 - Integration by substitution
 - Reduction formulae
 - Integration by trigonometric substitution
 - Integration using partial fraction decomposition
9.
 - Tests for convergence
 - Geometric and p series
 - Radius of convergence
 - Differentiating power series

Introduction to Programming

Exam Layout

2 hours long

Covered in Class

Number Systems

- Binary representations
- Converting integers to binary
- Converting fractions to binary
- Octal representations
- Hexadecimal representations
- Binary to octal/ hexadecimal
- Binary to decimal (Horners Method)

Computer Representations of Numbers

- Fixed-point
- Floating-point
- Conversions to floating-point numbers
- Disadvantages of floating-point numbers

Basic Programming

- Basics
- Loops
- Conditionals
- Functions
- Recursion
- Pseudocode
- Structures

Logic

- AND, OR, NOT

- Boolean expressions

- Truth tables

Proof Writing

- Structures of proofs
- Methods of proofs

Computational Work

- Operation counts
- o-notations

Covered Algorithms

- Estimate square roots (cube)
- Pi approximations
- Binary search (n-ary)
- Bisection root-finding
- Netwon's method
- Merge sort

Linear Algebra

- Vector-Vector multiplication
- Matrix-Vector multiplication
- Matrix-Matrix multiplication
- Matrix structures
- Column/ row major
- Sparse matrices
- Strassen's method

Introduction to Probability

Exam Layout

2 hours long, six questions

Questions will be quite similar to tutorial Questions

Just look at them :)